

Ash L. Gilbert

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EDUCATION

Ph.D. (Atmospheric and Oceanic Sciences) at University of Colorado-Boulder	2022 – 2027
M.S. (Atmospheric and Oceanic Sciences) at University of Colorado-Boulder	2022 – 2025
B.S.E., <i>summa cum laude</i> (Climate and Meteorology, conc. Meteorology) at University of Michigan	2018 – 2022
Minors in Computer Science and Latin	

AWARDS & HONORS

ATOC Student Service Award , University of Colorado-Boulder	2025
CIRES Graduate Student Research Award , University of Colorado-Boulder	2025 – 2026
Best Poster in Aerosols, Clouds, and Precipitation , ESSS Conference, University of Colorado-Boulder	2023
NSF Graduate Research Fellowship , University of Colorado-Boulder	2022 – 2025
James B. Angell Scholar , University of Michigan	2020 – 2022
Engineering Honors Program , University of Michigan	2020 – 2022
William J. Branstrom Freshman Prize , University of Michigan	2019
Dean's List , University of Michigan	2018 – 2022

RESEARCH INTERESTS

- polar cloud processes
- Community Earth System Model (CESM)
- Arctic climate
- wind nudging
- climate model Development
- internal climate variability
- climate change attribution
- atmosphere-sea ice interactions
- wildfire-climate interactions

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Colorado-Boulder 2022 – Present

Advisor: Professor Jennifer Kay

Project 1 (2023 – 2025): **A novel model hierarchy isolates the limited effect of supercooled liquid cloud optics on infrared radiation**

- Collaborator: Dr. Penny Rowe
- Evaluated the impact of supercooled liquid water optical properties on the Arctic climate via a hierarchy of climate models (CESM), ordered by level of dynamical constraint
- Identified a substantial effect from the optics in constrained dynamical models, but not in dynamically unconstrained models
- Proposed this model hierarchy as a method to assess the effect of model physics changes on a model climate
- First-author manuscript of this project has been published

Project 2 (2024 – 2025): **Observed winds alone cannot explain recent Arctic warming and sea ice loss**

- Collaborators: Professor Edward Blanchard-Wrigglesworth, Dr. Marika Holland, Dr. David Bailey, Professor Alexandra Jahn, and Dr. David Schneider
- Assessed the contribution of winds to recent Arctic warming and sea ice loss using wind nudging in CESM and comparing to observations and the pre-industrial CESM control
- Evaluated the change in wind contribution for an increase and decrease in mean state sea ice thickness
- Developed a tuned version of CESM2 that has a pre-industrial climate and modern-day mean state sea ice
- First-author manuscript of this project has been published

Project 3 (2025 – Present): **Internal and anthropogenic contributions to recent (1950-2024) Arctic temperature and sea ice change**

- Collaborators: Professor Edward Blanchard-Wrigglesworth
- Assessed the contribution of winds, greenhouse gases, and aerosols to recent Arctic warming and sea ice loss using wind nudging in CESM and single-forcing attribution framework
- Evaluated the performance of new wind nudging runs against observations and large ensemble runs

Research Assistant, University of Michigan
Project: **Watersheds, $\delta^{18}\text{O}$, and Climate of the Peruvian Andes**
Advisor: Dr. Phoebe Aron

2018 – 2019

- Interpolated meteorological, hydrological, and isotopic data with ArcGIS tools
- Developed and evaluated multiple procedures for calculating statistics of watersheds with ArcGIS tools
- Maintained a database of meteorological and isotopic information with SQL for Southern Peru
- Performed linear regression analysis on reanalysis datasets and isotope data using MATLAB
- Assisted PhD student relating to their study of hydrology in the Peruvian Andes by completing a variety of small research projects

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Colorado-Boulder Fall 2026
Climate Modeling Lab (ATOC 4870/5870), Professor William Chapman
Planning to give a lecture in November 2026

Instructional Aide, University of Michigan Fall 2021
Earth and Space System Evolution (CLIMATE 320), Professor Eric Kort

Collaborator, University of Michigan Fall 2019 – Spring 2020
Melting Ice Rising Seas Teach-Out

SERVICE

Journal Reviewer
Geophysical Research Letters

ATOC REU Mentor, University of Colorado-Boulder 2024

- Mentored an undergraduate student for a summer research project
- Gave technical and career advice

RECCS Mentor, University of Colorado-Boulder 2023

- Co-mentored an undergraduate student for a summer research project with another graduate student
- Gave technical and career advice

ATOC RSO, University of Colorado-Boulder 2024 – Present

- Helped organize and fund winter retreat and department SWAG

Prospective Student Committee, University of Colorado-Boulder 2023 – Present

- Helped organized prospective student visits in the spring

Engineering Honors Graduate School Panel, University of Michigan 2023 – 2025

- Served on five graduate school informational panels for seniors in the Engineering Honors program
- Answered questions about the graduate school application process and graduate school life

Graduate Application Mentorship Program Co-lead, University of Colorado-Boulder 2022 – Present

- Mentored several students each year applying to the department by giving advice and reviewing personal statements
- Since 2023, co-led program recruiting mentors and matching mentors with mentees

Technology Committee, University of Colorado-Boulder 2022 – 2024

- Made improvements to department website including removal of old links and addition of photos and personal website links to graduate student pages
- Advocated for committee to have department website access to make regular updates

Executive Board Member for Out in STEM, University of Michigan 2020 – 2022

- Collaborated with other board members to plan a large event for welcome week at the University of Michigan
- Led and organized several weekly meetings and discussions
- Created online pen pal program to connect members with each other outside of weekly meetings to make up for the lack of in-person networking during COVID-19 pandemic

COMPUTER SKILLS

- Python
- Jupyter
- C++
- SQL
- MATLAB
- Fortran90
- Community Earth System Model (CESM)
- ArcGIS
- Unified Forecast System (UFS)
- Autodesk Inventor
- Adobe Illustrator
- Microsoft PowerPoint
- LaTeX
- GitHub
- High-Performance Computing (HPC)
- Message-Passing Interface (MPI)
- OpenMP
- OpenACC

LANGUAGES

Latin (reading knowledge)

PROFESSIONAL ASSOCIATIONS

American Geophysical Union
American Meteorological Society
Out in STEM

PUBLICATIONS

Submitted

Gilbert, A.L., Kay, J.E., and Blanchard-Wrigglesworth, E. Internal and anthropogenic contributions to recent (1950-2024) Arctic temperature and sea ice change. *Environmental Research: Climate*. Submitted.

In Review

Wright, D.M., Jablonowski, C., Fujisaki-Manome, A., Mroczka, B., **Gilbert, A.**, Titze, D., Mann, G.E., and Anderson, E.J. The Sensitivity of Lake-Effect Snowfall to Changes in Lake Surface Conditions Across the Forecast Horizon in the Unified Forecast System's Short-Range Weather Application (UFS-SRW). *Monthly Weather Review*. In review.

Gale, S., Fu, Q., Wang, H., Blanchard-Wrigglesworth, E., Wang, M., **Gilbert, A.**, Sweeney, A.J., and Po-Chedley, S. Why April Stands Out: Monthly Impacts of Internal Variability on Arctic Amplification. *Journal of Geophysical Research: Atmospheres*. [[Preprint](#)]

Published

Gilbert, A.L., Kay, J.E., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D.P. (2025). Observed winds alone cannot explain recent Arctic warming and sea ice loss. *Environmental Research: Climate* **4**(4), 045009. [[PDF](#)]

Gilbert, A., Kay, J.E., and Rowe, P. (2025). A novel model hierarchy isolates the limited effect of supercooled liquid cloud optics on infrared radiation. *Geoscientific Model Development* **18**(19), 7185–7197. [[PDF](#)]

Jonko, A., Oliveto, J., Beaty, T., Atchley, A., Battaglia, M.A., Dickinson, M.R., **Gilbert, A.**, Godwin, D., Kupfer, J.A., Hiers, J.K., Hoffman, C., North, M., Restaino, J., Sieg, C., and Skowronski, N. (2024). How will future climate change impact prescribed fire across the contiguous United States? *npj Climate and Atmospheric Science* **7**(1), 1–10. [[PDF](#)]

Snide, C.E., **Gilbert, L.**, Meyer, A., Samson, P., Flanner, M., and Bassis, J. (2020). Seeing the Greenland Ice Sheet through students' eyes. *Eos*, 101. [[PDF](#)]

PRESENTATIONS

Seminars

Gilbert, A. Influence of winds and sea ice thickness on recent Arctic warming and sea ice loss. UCLouvain Earth and Life Institute: Climate. Louvain-la-Neuve, BE. June 2026.

Gilbert, A., Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Observed Winds Alone Cannot Explain Recent Arctic Warming and Sea Ice Loss. Atmospheric and Oceanic Sciences Colloquium.

Boulder, CO. September 2025.

Talks

- Gilbert, A.**, Kay, J., and Blanchard-Wrigglesworth, E. Attribution of Recent Arctic Warming and Sea Ice Loss in the Context of the Observed Winds. Community Earth System Model (CESM) Workshop. UCAR, Boulder, CO. June 2026.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Observed Winds Alone Cannot Explain Recent Arctic Warming and Sea Ice Loss. Climate and Cryosphere (CliC) Conference. Wellington, NZ. February 2026.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Community Earth System Model (CESM) Workshop. UCAR, Boulder, CO. June 2025.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. AMS Denver Summit 2025. Denver, CO. May 2025.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., Schneider, D. What controls the rapid Arctic warming since 1980? International Conference on Arctic Research Planning (ICARP IV) at Arctic Science Summit Week 2025. Boulder, CO. March 2025.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Land Ice and Polar Climate Working Group Meeting 2025. NCAR, Boulder, CO. March 2025.
- Gilbert, A.**, Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Polar Radiant Energy in the Far InfraRed Experiment (PREFIRE) Science Team Meeting. Boulder, CO. August 2024.
- Gilbert, A.**, Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Community Earth System Model (CESM) Workshop. UCAR, Boulder, CO. June 2024.
- Gilbert, A.** and Kay, J. A Breakdown of Arctic Warming. Land Ice and Polar Climate Working Group Meeting 2024. NCAR, Boulder, CO. February 2024.

Posters

- Gilbert, A.**, Kay, J., and Chepfer, H. An Observational Constraint on Future Greenland Precipitation Based on New Data from EarthCARE. AMS Madison Summit. Madison, WI. August 2026.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., Bailey, D.A., Holland, M.M., Jahn, A., and Schneider, D. Observed Winds Alone Cannot Explain Recent Arctic Warming and Sea Ice Loss. Earth System and Space Science (ESSS) Conference. Boulder, CO. December 2025.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., and Schneider, D. What controls the rapid Arctic warming since 1980? American Geophysical Union (AGU) Abstracts. A33B-1975. Washington, DC. December 2024.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., and Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Earth System and Space Science (ESSS) Conference. Boulder, CO. December 2024.
- Gilbert, A.**, Kay, J., Blanchard-Wrigglesworth, E., and Schneider, D. Isolating the Contribution of Observed Winds to Recent Arctic Warming and Sea Ice Loss. Graduate Climate Conference (GCC). Pack Forest, Eatonville, WA. November 2024.
- Gilbert, A.**, Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Cloud Feedback Modeling Intercomparison Project (CFMIP) Meeting. Boston College, Boston, MA. June 2024.
- Gilbert, A.**, Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Polar Amplification of Climate Change Across Hemispheres and Seasons: Causes and Constraints Workshop. UCAR, Boulder, CO. January 2024.
- Gilbert, A.**, Kay, J., and Rowe, P. Isolating the Influence of Temperature-dependent Cloud Optics on Infrared Radiation within a Model Hierarchy. Earth System and Space Science (ESSS) Conference. Boulder, CO. December 2023.
- Gilbert, A.** and Kay, J. Impact of Temperature Dependent Cloud Optical Properties on Modeled Infrared Radiation. Gordon Research Conferences (GRC) 2023 Radiation and Climate. Bates College, Lewiston, ME. July 2023.

- Gilbert, L.** and Bassis, J. Observing and Modeling Drainage Networks from Supraglacial Lakes on Russell Glacier, West Greenland. American Geophysical Union (AGU) Abstracts. C151C-0809. Virtual. December 2021.
- Gilbert, L.**, Rowe, P., Fergoda, M., and Neshyba, S. Influence of Temperature Dependence of Cloud Optical Properties on Simulated IR Flux at South Pole, Antarctica. American Meteorological Society (AMS) Abstracts. 136. Virtual. January 2021.
- Gilbert, L.** and Bassis, J. Observing and Modeling Drainage Networks from Supraglacial Lakes on Russell Glacier, West Greenland. American Geophysical Union (AGU) Abstracts. C061-0011. Virtual. December 2020.
- Gilbert, L.**, Aron, P., Samson, P., and Poulsen, C. Ensemble Probability Program for Backward Trajectories from the University of Michigan Biological Station (UMBS) based on the Relative Abundance of ^{18}O . Michigan Geophysical Union (MGU) Symposium. Ann Arbor, MI. April 2019.